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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 12

Team No.: 23

Project Title: **Multithreaded Linux Application Using POSIX Threads and Mutexes**

Team Members

Roll Number	Student Name	Signature
2520090032	D.MANYA AKSHITA	
2520090035	KIRAN SWAMY	
2520090144	P.NIRVIGHNA SAI	

2. Abstract:

The project “Multithreaded Linux Application Using POSIX Threads and Mutexes” focuses on the development of a Linux-based application that demonstrates concurrent execution and thread synchronization. The application uses POSIX Threads (Pthreads) to create and manage multiple threads, allowing different tasks to execute concurrently and improving the efficiency of program execution. Since multiple threads may access shared resources simultaneously, mutexes are implemented to provide mutual exclusion and prevent race conditions and data inconsistency. The project demonstrates important concepts of Linux multithreading, including thread creation, thread execution, synchronization, shared resource management, and thread termination. This application provides a practical understanding of how multithreaded programs can be designed and safely executed in a Linux environment using standard POSIX threading mechanisms.

1. Problem Statement:

Traditional single-threaded applications execute tasks sequentially, which can lead to inefficient CPU utilization and increased execution time when multiple tasks need to be performed. In addition, when multiple threads access shared resources simultaneously, race conditions and data inconsistency may occur. Therefore, there is a need to develop a Linux application that can execute multiple tasks concurrently while safely managing shared resources. The project addresses this problem by using POSIX threads for concurrent execution and mutexes for synchronization and mutual exclusion**.**

2. Objectives:

1. To understand multithreading and concurrent execution in a Linux environment.
2. To create and manage multiple threads using the POSIX Threads (Pthreads) library.
3. To execute multiple tasks concurrently to improve application efficiency.
4. To identify problems caused by shared resources, such as race conditions.
5. To implement mutexes for synchronizing access to shared resources.
6. To prevent data inconsistency when multiple threads access the same resource.
7. To understand thread synchronization and termination in Linux applications.
8. To gain practical experience in developing and executing multithreaded C applications on Linux.

3. Proposed Methodology

The project follows these main steps:

1. Create multiple threads using POSIX functions.
2. Assign different tasks to each thread for concurrent execution.
3. Identify shared resources accessed by multiple threads.
4. Implement mutex locks to control access to shared resources.
5. Execute and synchronize threads safely.
6. Analyze the results to verify correct execution and prevent race conditions.

5. Operating Systems Concepts / Linux APIs Used

(List the relevant Operating Systems concepts, Linux system calls, POSIX APIs, or mechanisms that will be used.)

OS Concept / Linux API / System Call	Purpose in the Project
Posix Threads	Used to create and manage multiple threads for concurrent execution.
Pthread_ create()	Creates individual threads to perform different tasks simultaneously.
Mutex (pthread_mutex_t)	Provides mutual exclusion when multiple threads access shared resources.

6. Individual Contribution

(Clearly specify the responsibility of each team member in the project.)

Roll Number	Student Name	Individual Responsibility
2520090032	D. Manya Akshita	POSIX thread creation and management using pthread_create() and pthread_join().
2520090144	P.Nirvighna Sai	Application testing, execution, result analysis, and project documentation.
2520090035	Kiran Swami	Mutex implementation and synchronization of shared resources.

7. Tools / Platforms / Software Used

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system environment for developing and executing the multithreaded application.
C / C++	Programming language used to implement the application and multithreading concepts.
POSIX Threads	Library used to create and manage multiple threads.
GCC Compiler	Used to compile the C/C++ source code and create the executable program.
Mutex / POSIX Synchronization	Used to synchronize threads and protect shared resources from race conditions.

8.Expected Outcome:

1. **Achieve concurrent execution of multiple tasks within the Linux application.**
2. **Use mutexes to safely manage access to shared resources.**
3. **Prevent race conditions and data inconsistency during concurrent execution.**
4. **Demonstrate proper thread synchronization and termination.**
5. **Improve understanding of Linux multithreading and POSIX APIs through practical implementation.**
6. **Successfully compile and execute the application in a Linux/Ubuntu environment.**

9. GitHub Repository

GitHub Repository Link: https://github.com/manyaakshitha07/KLH-CSIT-2026-17-Multithreading

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: Raghupathi M

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____